

JAYRALDINE'S CATERING MANAGEMENT SYSTEM

Formal Project Proposal & Functional Specification Document

1. Project Background & Business Case

Client: Jayraldine's Catering Services (Cebu, Philippines) | **Classification:** Integrated Catering ERP & Kiosk System

System Scope: Dual-Platform Solution (Windows PC Back-Office Suite + 100% Offline Tablet/Mobile Field Kiosk)

Project Goal: Automate and modernize end-to-end catering operations — replacing manual paperwork with automated client ordering, live financial estimations, downpayment enforcement, kitchen preparation workflows, expense tracking, AI business forecasting, and comprehensive reporting.

2. Operational Problems vs. Proposed System Solutions

Current Manual Operational Challenge	Proposed System Solution
Manual Headcount & Math Calculation Errors	Automated mathematical engine computing exact base totals, per-pax rates, custom add-ons, downpayment percentages, and balance ledgers without human error.
Unmonitored Expenses & Blind Profit Margins	Integrated Cost of Goods Sold (COGS), labor, transport, and utility expense tracking deducting directly from gross revenue to calculate true Net Profit.
Double Bookings & Venue Capacity Overload	Automated event date conflict detection and daily capacity hard-blocking (e.g. max 600 pax/day).
Remote Venue Disconnectivity	100% Offline-First architecture running self-contained SQLite engines on local hardware with zero internet dependency.
Consultation & Ordering Bottlenecks	Self-service guided touch ordering flow reducing client consultation time from 45 minutes to under 5 minutes.

3. Proposal Topic Breakdown for 3 Presenters

PRESENTER 1: Project Scope, System Architecture & Executive Oversight

- **Dual-Platform System Architecture:** Connects a Windows PC back-office workstation (Catering_Present) with a portable, 100% offline tablet/mobile kiosk (Tablet_PWA/APK) with 2-way master Excel and SQLite synchronization.
- **Executive Dashboard & Business KPIs:** Live monitoring of Gross Revenue, Total Operating Expenses, Net Profit Margins, and booking pipeline stages.
- **Capacity Governance & Protection:** Automated capacity validation preventing overbooking beyond daily kitchen production limits (e.g. max 600 pax/day).
- **Customer CRM & Loyalty Engine:** Client booking history tracking with automated loyalty tiers (Bronze, Silver, Gold, VIP) and scheduled follow-up reminder logs.

PRESENTER 2: Client Ordering Subsystem, Sales Workflows & Billing Automation

- **Guided 6-Step Client Ordering Engine:** Client lookup & split Cebu Barangay/Street address (Step 1); touch occasion picker & Package Details Viewer (Step 2); categorized dish selection with sticky filter header (Step 3); one-tap upsell add-ons like Lechon platters (Step 4); downpayment recording & legal terms (Step 5); pre-booking confirmation drawer & automated offline PDF receipt (Step 6).
- **Billing, Invoicing & Payment Records:** Official commercial invoice generation (INV-XXXX), payment milestone tracking, and partial balance ledgers.
- **Multi-Channel Notification Gateway:** Automated booking confirmation and PDF invoice dispatch via Email (SMTP) and SMS (Semaphore API).
- **Downpayment Enforcement:** Configurable threshold (e.g. 30%) blocking event confirmation without verified deposit.

PRESENTER 3: Financial Management, Expense Tracking, AI Forecasting & Kitchen Operations

- **Cash Flow & Expense Tracking:** Real-time Cash In / Cash Out monitoring, categorized expense logging (Food Costs, Labor Wages, Logistics, Utilities, Equipment), and automated Net Profit deduction (Revenue - Expenses = Net Profit).
- **AI Intelligence & Demand Forecasting:** AI forecasting of seasonal booking demand, smart raw ingredient portion estimator per pax count, and menu profitability analytics.
- **Kitchen Operations (Kanban Workflow):** Real-time kitchen order pipeline (Pending, Cooking, Plating, Ready for Transport, Dispatched) and interactive per-dish task checklists ensuring recipe quality.
- **Reports & Analytics:** Monthly/Annual revenue comparisons, top packages report, popular dishes breakdown, and payment method distribution charts.
- **Audit Governance & Security:** Complete change audit log recording actor, action, timestamp, and JSON data states with 100% offline data privacy.

4. Complete System Functional Module Matrix

Module	Core Functionality	Platform
Executive Dashboard	Real-time KPIs, Revenue, Net Profit, Capacity alerts, Recent Activity	PC Back-Office
Booking & Kiosk Wizard	6-step guided ordering, package selection, downpayment calculation	PC + Tablet Kiosk
Calendar & Scheduling	Visual event timeline, date conflict detection, daily capacity load	PC Back-Office
Menu & Package Catalog	CRUD packages, pax price matrix, dish categories, photo management	PC + Tablet Kiosk
Billing & Invoicing	Invoice generation, downpayment enforcement, balance collection	PC + Tablet Kiosk
Receipt & Notifications	PDF receipt generation, SMTP email dispatcher, SMS gateway	PC + Tablet (PDF)
Cash Flow Tracker	Inflow/Outflow tracking, payment method breakdown, net balance	PC Back-Office
Expense & COGS Manager	Food costs, labor wages, logistics, utilities, Net Profit margin	PC Back-Office
AI Demand & Forecaster	Pax forecasting, ingredient quantity estimator, menu insights	PC Back-Office
Kitchen Kanban & Checklist	Visual preparation cards, per-dish task checklist, order dispatch	PC Back-Office
Customer CRM & Loyalty	Booking history, automated loyalty tiers (Bronze to VIP), follow-ups	PC + Tablet Kiosk
Reports & Analytics	Revenue charts, package sales breakdown, PDF/Excel export	PC Back-Office
Security, Audit & Backups	Full change audit logs, 1-click DB backup/restore, Excel data sync	PC + Tablet Kiosk

5. Technical Specifications & Architecture

- **Back-Office PC System:** Native Python/Qt Desktop Application compiled to Windows executable (.exe).
- **Tablet Kiosk App:** HTML5 / JavaScript / CSS + Client-Side SQLite WASM + Standalone Android APK (7.1 MB).
- **100% Offline Capability:** Self-contained database engine; zero internet connection required for ordering, querying, or receipt printing.
- **Data Security & Ownership:** Stored locally on physical business hardware; zero third-party cloud data leaks or monthly subscription fees.